

Problem 1

1A : Feature Construction Description

We did not apply explicit normalization or standardization to the input features. Instead, we performed preprocessing steps including imputing missing values (filling age with the training set median and other missing values with 0), converting binary categorical variables to 0/1 indicators, and one-hot encoding multi-class categorical variables such as background_father, background_mother, and region. We chose not to normalize features because our final model is a Random Forest, which is not sensitive to feature scaling. Additionally, we used class_weight='balanced' to account for class imbalance in the training data.

We created two types of image features, color variation and symmetry. For color variation, we computed the standard deviation of pixel intensities for each RGB channel (r_std, g_std, b_std), which captures how much the color varies across the lesion. We decided to implement this because malignant lesions often have more irregular pigmentation, which we would now be able to detect. For symmetry, we computed both vertical and horizontal symmetry by comparing each half of the image to its flipped counterpart. This captures the tendency of malignant lesions to be more asymmetric.

We also experimented with undersampling the majority class, but found that the undersampling significantly reduced the training data. We found that each class would have around 30 samples, which hurt the model's performance. As a result, we instead used class weighting to handle imbalance.

We created a total of 5 image-derived features. These 5 features include three color variation features (r_std, g_std, b_std) and two symmetry features (vertical and horizontal symmetry).

We used the entire image to construct our features. The color variation feature is computed over all pixels in the image, and the symmetry features compare halves of the full image. Although the symmetry calculation splits the image into regions, it still utilizes information from the entire image rather than focusing on a zoomed in region.

1B : Cross Validation Design Description

The performance metric we optimized on held-out data was AUROC. This metric is appropriate because it measures the model's ability to distinguish between positive and negative classes and is robust to class imbalance in the dataset.

We used 5-fold cross-validation and sampled 10 random hyperparameter configurations using RandomizedSearchCV (n_iter = 10), resulting in a total of 50 model fits. In each split, approximately 80% of the data was used for training and 20% for validation.

After cross-validation, we selected the hyperparameter configuration with the highest average AUROC across folds. We then trained a final Random Forest model using these optimal parameters on the full training dataset and used this model to generate predictions on the test set.

We used off the shelf libraries for implementation, specifically scikit-learn for modeling and cross-validation, along with matplotlib and seaborn for visualization.

1C : Hyperparameter Selection for Random Forest Classifier

We searched over a grid of hyperparameters that control the complexity and capacity of the Random Forest. Specifically, we varied `max_features = {1, 2, 3}`, `max_depth = {8, 16, 32, 64, 128}`, `min_samples_leaf = {1, 5, 10}`, and `n_estimators = {10, 100, 500, 1000}`.

We fixed several hyperparameters to non-default values, including setting `class_weight = 'balanced'` to address class imbalance and `random_state = 1` for reproducibility. We also set `criterion = 'entropy'` instead of the default `gini`, as entropy uses information gain to measure the quality of splits and can sometimes provide more informative partitions when classes are imbalanced. Finally, we fixed `min_samples_split = 2` to allow trees to grow fully, relying on `max_depth` and `min_samples_leaf` to control overfitting.

Random Forests are well suited for this problem because they are robust to noisy predictors and can naturally handle both tabular and engineered image features. Their ensemble structure reduces variance, which is helpful given the class imbalance.

The hyperparameters we varied directly control model complexity. Small `max_depth` or large `min_samples_leaf` strongly regularize the trees and can lead to underfitting, while deeper trees with small leaves allow the model to capture more detailed structure at the risk of overfitting. Random Forests do not involve gradient-based optimization, so there are no step-size or convergence issues. Instead, we control overfitting structurally through these hyperparameters.

To generate the figure, we fixed `max_features = 2`, `min_samples_leaf = 1`, and `n_estimators = 100`, and varied `max_depth`. The plot shows both training and validation AUROC, along with error bars representing the standard deviation across folds and individual fold scores.

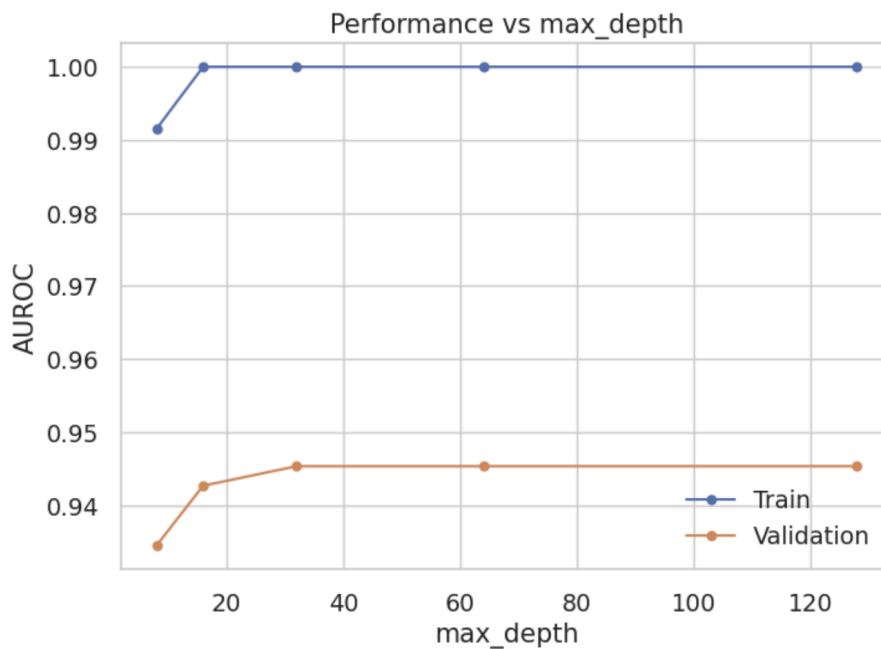


Figure 1: Validation AUROC vs max_depth for Random Forest

From our figure, we observed that the validation AUROC increases from max_depth = 8 to 16 and reaches its highest mean value at max_depth = 32. The depths 32, 64, and 128 all achieve similar mean AUROC values (0.945), indicating that performance plateaus once the trees are moderately deep. Although the training AUROC reaches 1.0 for depths greater than 16 (indicating overfitting at the individual tree level), validation performance remains stable due to the variance-reducing effect of the ensemble. This suggests that max_depth = 32 is a reasonable choice, though the evidence is not strongly decisive since several larger depths perform similarly.

1D : Reflecting on Your Test Set Performance

From Gradescope, our model achieved a test AUROC of 0.90755, which is slightly lower than our best cross-validated validation AUROC of approximately 0.945. The model performed extremely well on sensitivity (TPR = 1.0) but struggled with specificity (TNR = 0.17), indicating that it correctly identified all malignant lesions but misclassified many benign ones as positive.

This imbalance likely reflects the model prioritizing recall of the malignant (minority) class, which leads to more false positives. Despite this, the high AUROC and AUPRC indicate that the ranking quality of the classifier is strong.

In the future, we could improve performance by tuning the decision threshold to better balance sensitivity and specificity, as well as by incorporating additional tabular and image features into the Random Forest model.

2A : Feature Representation description

We tried separating the data set into 3 separate data sets, for young, middle, and old, believing that we could train each data set and attain a high accuracy for each by separating by age group. Then, we created 3 separate neural networks to train on each of the data sets. However, when we examined the data shape after, we found that the size of the old data set was highly overbalanced in comparison to the young data set. The senior class especially performed poorly. After trying one neural network, we used a CNN, thinking that it would help because now the model would look directly on the images rather than just the table data. However, the CNN only managed to barely improve our AUROC. Then, we used the neural network with the complete data set, but it also did not improve our accuracy much. Thus, we decided to use Bayesian optimization combined with a random forest classifier, which we will expand on below.

Originally we felt like the same two features from 1A were enough. However, we decided to add in the border difference feature, because color and symmetry were not enough to differentiate between older vs younger patients. Color can vary a lot with age, and in small lesions, symmetry may not be able to accurately capture whether lesions were cancerous or not. The border differentiation in particular would give us more detail into whether the model would be able to accurately spot the difference between the cancerous skin lesions (uneven, darker edges) and the non cancerous skin lesions (smoother, more uniform in color).

2B : Cross Validation (or Equivalent) description

Like 1B, we optimized AUROC. This metric is appropriate because it measures the model's ability to distinguish between positive and negative classes and is robust to class imbalance in the dataset.

However, instead of using RandomizedSearchCV, we used 5-fold StratifiedKFold cross-validation, fitting the classifier with sample weights. We chose StratifiedKFold because stratification preserves the class proportions in every train/validation split, which is crucial when the dataset is imbalanced. This helps prevent some folds containing little to no minority class data, which could lead to unstable training and inflated variance/ AUROC estimates.

After cross-validation, we selected the hyperparameter configuration with the highest average AUROC across folds. Then, we used Bayesian optimization combined with the random forest model, to train using these optimal parameters on the full training dataset. Bayesian

optimization chooses the next set of hyperparameters based on past results. We then used this model to generate predictions on the test set. We used Bayesian optimization with the model because it explores the hyperparameter space more efficiently than a random or grid search.

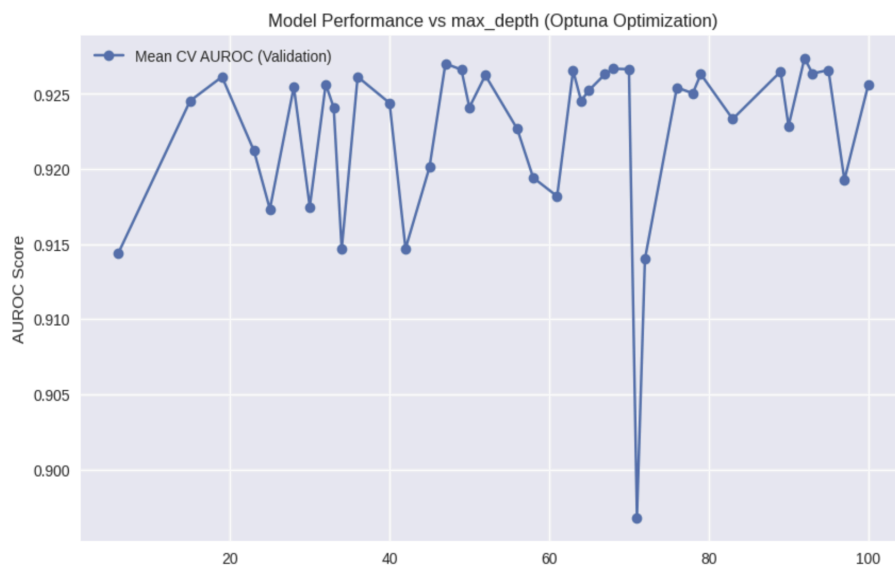
2C : Classifier description with Hyperparameter search

Like 1C, we searched over a grid of hyperparameters that control the complexity and capacity of the Random Forest classifier. Specifically, we tested a range between $\{10, 1000\}$ for `n_estimators`, a range between $\{5, 100\}$ for `max_depth`, a range between $\{1, 4\}$ for `min_samples_leaf`, between $[\text{sqrt}, \log 2]$ for `max_features`, and setting criterion to entropy.

We fixed several hyperparameters to non-default values, including `random_state = 1` for reproducibility. However, we instead explicitly gave our old, middle, and young classes weights to balance the class out more. We overweighted the young class, gave a neutral weight to the middle class, and slightly underweighted the old class.

Random Forests are well suited for this problem because they are robust to noisy predictors and can naturally handle both tabular and engineered image features. Their ensemble structure reduces variance, which is helpful given the class imbalance. Combined with Bayesian optimization's efficiency at exploration, we hoped this combination would find the hyperparameters that would lead to the highest accuracy.

To generate the figure, we fixed `max_features = log2`, `min_samples_leaf = 1`, and `n_estimators = 429`, and varied `max_depth`. The plot shows the mean validation AUROC.



From our figure, we observed that the validation AUROC learns the training data well, but never seems to improve beyond 0.925. This suggests that increasing the number of trees does not necessarily lead the model to overfit, but also does not allow the model to improve on learning the complexity of the features. Aside from one outlier around $\text{max_depth} = 70$, the graph suggests that the Bayesian optimization is simply exploring the hyperparameters without really committing to one direction or another. This has major implications for our ultimate accuracy on our test set, meaning that our model has not improved on our Problem 1 attempt, despite our attempts at addressing class imbalance and accuracy across all age groups.

2D : Reflecting on Your Model Performance

From Gradescope, our model achieved a test AUROC of 0.9065. However, the age-sectioned results showed that we really struggled with the Senior class. We obtained an AUROC of 1.0 for the “Young” group, 0.947 for “Adults,” but only 0.793 for the Senior group. Because the Young group is so under-sampled, the model may be learning a decision boundary that is overly influenced by the Adult and Senior groups, making it harder to generalize well across all ages. Increasing the Young-group representation could help create a clearer and more balanced boundary. Another plausible factor is that the Senior group may have more variability in how their data was collected or recorded. For example, their images or table data may be captured under less consistent conditions, leading to noisier inputs. Even small inconsistencies in image quality, lighting, or how information is entered can introduce additional variance that makes the model’s decision boundary harder to learn.

In the future, we could improve this performance by obtaining more data for the Young group, which currently has only 25 samples, or by standardizing the data collection process for the Senior group to reduce variability in image quality and table data.

Problem 2

